# ENSF-381: Full Stack Web Development Laboratory

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Lab

## Objectives

Welcome to the ENSF381 course lab! In this lab, we will focus on fundamental React concepts. Specifically, we will explore the fundamentals of React by practicing JSX, creating reusable components, and passing data using props. We will also learn how to render lists efficiently and improve code organization by modularizing components into separate files.

## Groups

Lab instructions must be followed in groups of **two students**.

## Constraint

Use only the JavaScript functions and data structures covered in lectures or in this lab session.

## Submission

- You must submit the complete source code, ensuring it can be executed without any modifications.
- If requested by the instructor, you may need to submit additional documentation (e.g., word documents and images).
- The link to the GitHub repository created for this lab must be provided alongside your submission on D2L. If you submit a word document as a part of this lab, include the link to the GitHub repository within that document as well.
- Only one member of the group needs to submit the assignment, but the submission must include the names and UCIDS of all group members.

## Deadline

Lab exercises must be submitted by 11:55 PM on the same day as the lab session. Submissions made within 24 hours after the deadline will receive a maximum of 50% of the mark. Submissions made beyond 24 hours will not be evaluated and will receive a grade of zero.

# Exercise 1: Understanding JSX Syntax in React

## Objective

In this exercise, you will practice using JSX, a syntax extension for JavaScript that allows you to write HTML-like code inside JavaScript. You will create a basic React component and experiment with embedding JavaScript inside JSX.

1. **Create a React App**
    - Open your terminal and create a new React app using the command:

```
npx create-react-app jsx-exercise
```

- Navigate to the jsx-exercise folder and open the project in your IDE.

2. **Create Your First JSX Element**
    - In the App component (src/App.js), replace the default code with a simple JSX element.
    - The element should include a heading (<h1>) that says "ENSF-381: Full Stack Web Development".
    - Below the heading, include a paragraph (<p>) with the text "React Components".
3. **Embed JavaScript Expressions in JSX**
    - Inside the App component, create a variable currentYear that holds the current year. Hint: use JavaScript's Date() method.
    - Create a new <p> element that displays currentYear inside the JSX by embedding the variable within curly braces {}.
4. **Using a Conditional Statement in JSX**
    - Create a boolean variable isLoggedIn and set it either to true or false.
    - Inside the JSX, conditionally render a message using a ternary operator. If isLoggedIn is true, display "Welcome back!"; otherwise, show "Please log in."

## Submission

1. Fill out the names and UCIDs of all group members in Answer_sheet.

2. Create a new GitHub repository and upload the code for Exercise 1 to the new repository. **NOTE: Do not include the node_modules folder.**

## Exercise 2: Component Composition

**Objective:** In this exercise, you will create a page similar to the user directory page from Lab 04. The goal is to practice breaking a page into smaller components and composing them together. In this version, the page remains completely static: no event handling, and no dynamic behavior are required.

### Instructions

1. Continue working with the same React app you created in the previous exercise.
2. Create a new file called Controls.js inside src. In this file, create a Delete component that renders a text box and a Delete button.
3. In the same Controls.js file, create a Controls component that renders the <Delete /> component followed by three additional buttons: Sort by Group, Sort by ID, and Grid / List. Export only Controls from Controls.js.
4. Create a UserList component in src/UserList.js.
    - Copy the provided users.js file into your React app's src folder
    - Import users into UserList.js
    - Use .map() to render each user
    - For each user, render the following structure:

```
<article>
  <h3>{user.first_name}</h3>
  <p>User Group: {user.user_group}</p>
```

```
  <p>ID: {user.id}</p>
</article>
```

5. Create a Footer component in src/Footer.js. Inside the component, use JavaScript's Date to get the current date, then render a simple <footer> that displays it alongside the first names of the members of your group.

6. In App.js, render the following content only when isLoggedIn is true:

   - Inside the conditional block, render one <section> containing <Controls />
   - Inside the conditional block, render one <section> containing the user list heading and <UserList />
   - Inside the conditional block, render <Footer />

7. Copy the provided App.css file into your React app's src folder, and ensure that App.css is imported into App.js so the styles are applied to your page.

## Submission

1. Upload the code for this exercise to the same repository you created for Exercise 1. **NOTE: Do not include the node_modules folder.**
2. Compress the updated React app and upload it to D2L along with the completed Answer_sheet.docx. **NOTE: Do not include the node_modules folder.**